

Received 11 March 2025, accepted 29 March 2025, date of publication 4 April 2025, date of current version 15 May 2025.

Digital Object Identifier 10.1109/ACCESS.2025.3558120

RESEARCH ARTICLE

A Comparative Analysis of Time Series Forecasting Models and the Novel ELSTMD Approach for Renewable Energy Generation

YUSEF ALKHANAFSEH¹, TAHIR CETIN AKINCI², (Senior Member, IEEE),
AND ALFREDO A. MARTINEZ-MORALES²

¹Electrical Engineering Department, Istanbul Technical University, 34469 Istanbul, Türkiye

²Bourns College of Engineering Center for Environmental Research and Technology, University of California Riverside, Riverside, CA 92521, USA

Corresponding authors: Tahir Cetin Akinci (tahircetin.akinci@ucr.edu) and Alfredo A. Martinez-Morales (alfredo.martinezmorales@ucr.edu)

This work was supported in part by California Energy Commission (CEC) under Grant EPC-16-077.

ABSTRACT The increasing number of people and the huge industrial evolution have led to serious environmental consequences, which have made energy from Renewable Energy Sources (RESs) inevitable. However, these energy sources suffer from intermittency as they depend on natural sources, which change over time. Consequently, this research addresses the issue by proposing an ANN model for reliable renewable energy forecasting. The Encoding Long Short-Time Memory Decoding (ELSTMD) model consists of two encodings, one LSTM, and two decoding layers. Moreover, the introduced model and 21 time series forecasting models are evaluated, including statistical, Artificial Intelligence (AI), and hybrid approaches. While making predictions, the impact of 17 distinct features, categorized into five primary groups—electricity, weather, economic, and seasonality-related attributes, as well as natural disasters—has been analyzed. Thus, the models are trained as univariate time series with exogenous variables. To evaluate the model's generalization, an additional dataset involving photovoltaic (PV) panels was used. The proposed model outperforms all competitors on both datasets, achieving an (R^2 , MAE, MSE, RMSE) of (0.56714, 0.06398, 0.00691, 0.08313) on the first dataset and (0.76574, 0.09344, 0.01417, 0.11903) on the second dataset. The most notable advantage of this research can be summarized as providing precise predictions for the electricity generated from RESs, which can help reduce the intermittency, randomness, and stability issues associated with these sources. It also plays a crucial role in optimizing electrical power systems' planning and storage efficiency.

INDEX TERMS Artificial neural network, decoding, encoding, forecasting, LSTM, renewable energy generation, time series analysis.

I. INTRODUCTION

Recently, electricity has become one of the most used forms of energy [1], encompassing various aspects such as generation, transmission, distribution, and market dynamics. Over the past few decades, electricity has become a fundamental part of our daily lives [2], [3], and the growth of emerging economies has dramatically increased its demand. This trend is expected to continue, with energy usage projected to double

from 2024 to 2040 [4]. Electricity generation can be achieved through several methods, including coal, natural gas, nuclear energy, and oil, which collectively account for three-fourths of total generation [5]. However, these sources pose significant environmental risks, as they are major contributors to greenhouse gas emissions, accounting for approximately 40% of Carbon Dioxide (CO₂) emissions [6], thus exacerbating the impacts of global warming [7]. Accordingly, there is a pressing need for new studies to explore alternative electricity generation sources. Renewable energy sources (RESs) such as solar, wind, geothermal, bioenergy, hydropower, and

The associate editor coordinating the review of this manuscript and approving it for publication was Akshay Kumar Saha¹.

ocean energy [8] have gained significant importance in the realm of electricity generation due to their status as environmentally friendly alternatives. These sources are crucial for reducing reliance on traditional fossil fuels, which are associated with high levels of pollution and greenhouse gas emissions [9]. The adoption of RESs is not only a matter of environmental concern but also of strategic importance for achieving energy security and sustainability. Furthermore, with the global shift towards renewable energy, it is anticipated that CO₂ emissions will decline dramatically, with projections indicating a reduction of more than 100% over the next 20 years [4]. This transition underscores the critical role of RESs in mitigating climate change and promoting cleaner electricity generation. Even though RESs are widely regarded as clean, sustainable, and naturally replenished in contrast to fossil fuels [10], they possess inherent limitations. These include unscheduled uncertainty, energy deficiency, reserve capacity requirements, volatility, intermittency, randomness, and stability issues, which stem from various factors such as the use of power electronics, unpredictable weather conditions, and temporal constraints [11], [12], [13]. Hence, forecasting the output and the availability of RESs can be crucial in reducing the previously stated limitations, which enhances the planning, management, and operation stages of electrical power [14]. Generally, the techniques used in forecasting renewable energy output are involved in four different categories, which are physical methods, statistical models, Artificial Intelligence (AI) techniques, and their hybrid methods [15].

Currently, with the huge interest in AI and its various aspects, such as Machine Learning (ML) and Deep Learning (DL), these technologies have become a trend across all industry layers, including the foundational, technological, and application levels [16]. The primary factors that have played a significant role in this are the increased number of AI papers, the substantial amount of AI venture capital, and the growing number of new AI startups [17]. In addition, AI has shown huge potential in various fields of electrical engineering, achieving accuracy levels that in some cases exceed 99% [18], from the generation step to the electricity market step. It enhances failure prediction, power generation, and control, Demand-Side Management (DSM), cyber security, customer churn prediction, diagnostic and maintenance tools, diagnostics and maintenance, enabling real-time monitoring, as well as the integration of Electric Vehicles (EVs)—for example, optimizing energy storage systems. References [19], [20], and [21]. AI models can be utilized in various electrical engineering applications, including classification, clustering, and forecasting.

In general, forecasting can be divided into three different horizons, which are short, medium, and long terms. They are limited differently based on the available dataset. Some researches categorized them as monthly, seasonal, and yearly [22]. Others categorized them as 1 to 6 months, 7 to 12 months, and more than 13 months, respectively [23]. However, other researchers divided them into four horizons, which

added a new one, a very short horizon. In this research, the horizons stated in [24] are found to be the most appropriate with the available data which states few seconds to 30 minutes for very short-term, 30 minutes to 6 hours for short-term, from 6 hours to 1 day medium-term, and from 1 day to 1 week ahead for long-term. Among other countries, Turkey can be considered one of the important countries that values RESs. Significant population, industrialization, and urbanization increases have transformed Turkey into a competitive market for several services [25], necessitating detailed analysis. According to the Republic of Turkey Ministry of Energy and Natural Resources, it is projected that approximately \$5.61 billion will be invested in the energy sector in 2024, with this number expected to reach \$20.2 billion by 2030 [26].

This study employs a novel AI-based time-series forecasting model, Encoding Long Short-Term Memory Decoding (ELSTMD), to generate long-horizon daily forecasts of electricity production from renewable energy sources (RESs) in Turkey. The collected dataset comprises 17 independent features categorized into five major groups: electricity, weather, economic, natural disaster, and seasonality-related features. Electricity-related features include electricity generated from fuel oil and natural gas, and maximum RES capacity. Weather features encompass cloud cover percentage, humidity, rain precipitation, pressure, snow, solar radiation, temperature, and wind speed. Economic features are RES unit cost and Gross Domestic Product (GDP). COVID-19 cases represent the natural disaster category. Additionally, two self-generated features, season and change points, relate to summer and winter seasons and abrupt changes in RES generation. To demonstrate the generalization of the proposed model, a supplementary dataset from photovoltaic (PV) systems in India was employed as well. This dataset comprises DC and AC power outputs, daily and cumulative inverter yields, ambient and module temperature measurements, and irradiation levels, and it is used to generate forecasts over a shorter time horizon. ELSTMD model, then, is compared to 21 different models in terms of 4 different metrics, which are R^2 , Mean Absolute Error (MAE), Mean Squared Error (MSE), and Root Mean Squared Error (RMSE). All the models are tuned using techniques such as Grid search, Random search, and trial-and-error to get the best possible results. These models can be classified into seven distinct categories:

- **Linear Models:**

- Automatic Relevance Determination Regression (ARDReg)
- Elastic Net Regression (ElasticNet)

- **Tree-based Models:**

- Random Forest Regressor (RFR)
- Extremely Randomized Trees Regressor (Extra-TreesReg)
- Extreme Gradient Boosting (XGBoost)
- LightGBM Regressor (LGBMReg)
- Gradient Boosting Regression Trees (GBRT)
- Histogram-Based Gradient Boosting Regressor (HGBR)

- CatBoost Regressor (CatBoostReg)
- **Ensemble Models:**
 - Voting Regressor (VotingReg)
 - Stacking Regressor (StackingReg)
- **Artificial Neural Network (ANN) Models:**
 - Temporal Convolutional Network (TCN)
 - Neural Basis Expansion Analysis Time Series (NBEATS)
 - Gated Recurrent Unit (GRU)
 - Long Short-Term Memory (LSTM)
- **Traditional Time Series Models:**
 - AutoRegressive Integrated Moving Average (ARIMA)
 - Seasonal AutoRegressive Integrated Moving- Average with eXogenous regressors (SARIMAX)
 - Facebook Prophet (FbProphet)
- **Kernel-Based Models:**
 - Support Vector Regression (SVR)
- **Hybrid models:**
 - LSTM and SARIMAX
 - LSTM and GRU

To provide a deeper understanding of the ELSTMD model, we meticulously detail its internal and overall structure, including a comprehensive layer-wise summary. We also address reproducibility—discussing hyperparameter settings, computing environment, and software versioning—as well as computational cost, scalability, and limitations. Furthermore, all implementation details are properly explained, and model interpretability is provided through an in-depth analysis of feature importance and an ablation study.

The organization of the paper is as follows: Section I introduces the research objectives and provides the overall context. Section II reviews previous and related work on the subject. Section III discusses renewable energy sources and their potential in Turkey. Section IV presents the background of the utilized methods, including their significance and underlying mathematical formulations. Section V explains the system's main architecture and data pipeline, while Section VI details the proposed ELSTMD model—including its internal structure, reproducibility, and discussions on computational cost, scalability, and limitations. Section VII provides an overview of the data collection process. Section VIII details the implementation process—including data preprocessing, feature engineering, and feature orthogonalization—and presents numerical results alongside an in-depth discussion. Finally, Section IX concludes the paper with a substantial summary and future research directions.

II. RELATED WORK

As RESs generally depend on fluctuating natural factors [27], their production is often considered intermittent, making the amount of energy they generate ambiguous. Therefore, forecasting RESs generation is a highly challenging task. To address this, numerous studies have been conducted, which can be broadly classified into two key categories: qualitative (non-numerical) and quantitative (numerical) [28].

The nonnumerical category depends on general experiences, attitudes, and behavior, while the numerical category uses mathematical approaches, which can generally be divided into four principal sub-categories: statistical methods, physical models, AI techniques, and hybrid approaches combining the previously mentioned methods [15]. In short, physical methods, such as Zephyr, Prediktor, and ANEMOS [29], depend on making Numerical Weather Predictions (NWP) for atmospheric dynamics based on physical parameters such as temperature, wind speed, pressure, etc [30]. However, they have some significant limitations, such as high computational cost [31] and insufficiency for short-term forecasting horizons [13]. On the other hand, statistical methods such as the Auto-Regressive Integrated Moving Average (ARIMA) [32] and Theta [33] try to find the mathematical relationship between variables. AI techniques, such as Recurrent Neural Networks (RNNs), can overfit the given data as they can fit non-linear relationships. When AI methods are compared to statistical methods, AI models outperform when there is an abundant quantity of data; when this condition is not satisfied, statistical methods perform better [34]. Eventually, hybrid techniques use a combination of these methods together.

The research [35] examined the performance of both ANNs and support vector regression in order to predict renewable energy production in Turkey. The period considered is between 1990 and 2020. Their data consists of two prime groups. The first one is energy structure which includes coal-based energy production, gas-based energy production, and liquid fuel (oil products)-based energy production and energy consumption per capita. The other group is a socio-economic and environmental group, which includes GDP (US Dollar), greenhouse gas emission per capita, and energy sector greenhouse gas emission variables relationship related to the CO₂ emission and population of the country. Finally, they choose geothermal, solar, wind, solid biomass, biogas, and waste (GWh) as dependent variables. The obtained values of (R^2 , RMSE, and MSE) for both ANNs and support vector regression are (0.92, 0.045, 0.002) and (0.899, 0.071, 0.005), respectively. Nevertheless, the training and validation processes of these models are not presented in the study, rendering it ambiguous to assess their results. For this reason, it is unclear whether the models suffer from issues such as overfitting or underfitting. Besides, the structure and hyperparameters of the ANN are not disclosed in the study.

The main goal of the study of [36] is to forecast of Turkey's total installed capacity of renewable and hydro energies and electricity generation from 2019 to 2030. For this purpose, they used the data from 2009 to 2018. The data for 2009–2017 was taken from the International Renewable Energy Agency (IRENA) database, and the data for 2018 year was taken from the Republic of Turkey Energy Market Regulatory Authority (EPDK). They summed values of hydro, wind, biomass, geothermal and solar energies as the data for total renewable energy. They used and compared the fractional nonlinear grey Bernoulli model (FANGBM) to its reduced forms. The evaluation is done based on MAPE metrics, which were

7.789, 7.300, 7.391, and 7.046 for GM, NGBM, FAGM, and FANGBM, respectively. As a result, the FANGBM achieved the highest accuracy. Despite that, the prediction results in the study exhibit an almost entirely linear increase, almost without any non-linearity. This indicates that the results may not be realistic enough to be considered reliable.

In [37], Fuzzy time series (FTS) model was used for predicting the renewable energy generation in Turkey. Specifically, they proposed a hybrid methodology that combines the intuitionistic FTS method with the intuitionistic fuzzy c-means. The included data history are the years between 2000–2020. The dataset used was collected from the Ministry of Energy and National Sources of Turkey's website on a yearly basis. They used MAPE and MSE as evaluation metrics, yielding values of 2.25 and 219,828, respectively, for the proposed model. The most noticeable disadvantage of this research is that the data used is significantly small, and the analysis of seasonality or trends has not been conducted.

The authors of [38] utilized five different models, namely ANN, LSTM, CNN, CNN + LSTM, and LightGBM, for short-term forecasting on an hourly basis of renewable energy production in a single city in Turkey, Sivas. The evaluation metrics used were RMSE and MAE. The models achieved the following RMSE and MAE values, respectively: 2.1617 and 1.4230, 2.1557 and 1.4975, 2.0354 and 1.4654, 2.1036 and 1.6028, and 1.7479 and 1.1072. They found that LightGBM performed the best in terms of these two metrics. Two types of data were used: renewable energy production and meteorological data. The former was obtained from Enerji Piyasaları İşletme Anonim Şirketi (EPIAS) and the latter from the Sivas Provincial General Directorate of Meteorology. The data covered the period from 1 January 2019 to 31 December 2020. However, it is apparent from the validation and training procedures that the established models suffer from overfitting issues, as indicated by the validation process starting from a very low point. Moreover, the LSTM model is particularly affected by overfitting, as evidenced by its significantly higher validation loss than its training loss.

The study by [39] introduced a GBRT model, which was initially developed for classification and later extended to regression tasks. Model selection and parameter tuning were performed using 5-fold cross-validation, and the feature importances were reported. The model is employed to forecast multi-site solar power generation for horizons ranging from one to six hours. The study utilized a dataset comprising 316 daily samples from 42 individual PV installations in Japan. For the six-hour horizon, the GBRT model achieved an RMSE between 0.100 and 0.137. However, unlike conventional time series methods, the model lacks a recursive updating mechanism when new observations become available, which is a notable limitation.

Another approach with significant potential for forecasting solar power generation is the hybrid model that integrates an AutoEncoder (AE) with a Long Short-Term Memory (LSTM) neural network, termed Auto-LSTM, as presented

in [40]. This model was evaluated using the GermanSolar-Farm dataset, which comprises data at a three-hour resolution over 990 days and is associated with 21 PV installations in Germany. For day-ahead forecasting, the model achieved an RMSE of 0.0713. However, the study exhibits several limitations, notably the absence of an ablation study, analysis of feature importance, discussion on the model hyperparameters, and a rigorous description of the training procedure.

In [41], the authors introduce a novel machine learning and statistical hybrid model (MLSHM) termed AutoEncoder GRU (Auto-GRU), designed specifically for forecasting PV power generation. The model was evaluated using two distinct datasets comprising meteorological data and PV power output. The first dataset, acquired from Shagaya in Kuwait, spans from May 5, 2017, to June 30, 2018, while the second dataset—recorded at five-minute intervals—originates from Cocoa Single Poly-SI in Florida, USA, covering the period from January 21, 2011, to March 4, 2012. The authors assert that the integration of statistical methods with machine learning models yields superior performance compared to using machine learning alone. Specifically, the model achieved normalized Mean Absolute Error (MAE) and normalized Mean Squared Error (MSE) values of 0.0394 and 0.00251 on the first dataset, and 0.0952 and 0.019 on the second dataset, respectively. However, the study is limited by its omission of information regarding the training procedure and other methodological aspects, keeping the reproducibility and clarity of the results ambiguous.

III. RENEWABLE ENERGY SOURCES AND POTENTIAL IN TURKEY

Geographically, Turkey has an almost rectangular shape, with area of 814,578 square kilometres. It is situated in the northern hemisphere, approximately midway between the equator and the North Pole, with coordinates ranging from 36 degrees N to 42 degrees N in longitude and 26 degrees E to 45 degrees E in latitude. Given its unique geographical position, Turkey can be regarded as a significant country in the context of RESs for electricity generation. Furthermore, substantial investment plans are being formulated, with an estimated \$5.61 billion allocated for 2024, projected to increase to \$20.2 billion by 2030. Currently, in Turkey, the most used RESs are wind, geothermal, reservoir, channel, river, Land-fill Gas (LFG), biogas, sun, and biomass. The amount of electricity generated from various RES in Megawatt hour (MWh) is shown in Figure 1 (a), covering the period from January 1, 2018, to March 31, 2024. It is evident that wind is the most and biogas is the least utilized RESs, and the use of solar energy (sun) has dramatically increased in recent years. Biomass has been steadily increasing, while the use of reservoir and channel-based RES has declined in the last two years. The percentage of electricity generated, in the same period, using these RESs is displayed in Figure 1 (b). The Figure 1 is based on data fetched from [42].

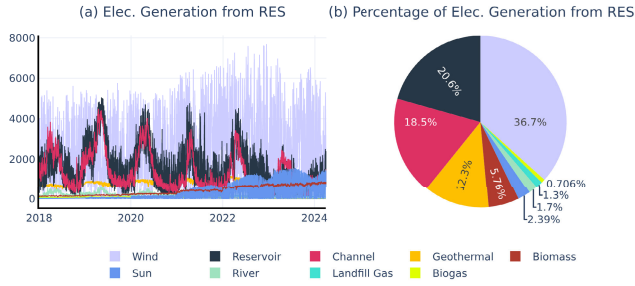


FIGURE 1. Potential of electricity generation from RESs in Turkey.

IV. BACKGROUND

During this research, several mathematical approaches have been utilized. This section aims to briefly explain each approach to ensure clarity. The approaches can be summarized into seven sub-sections, which are:

A. DATA IMPUTATION

The iterative imputation technique, as the concept itself shows that it is a process applied to handle missing data values iteratively [43]. Its working principle depends on two major steps: initialization and iteration. The first one is completely a guess value. Sometimes, the mean of the values is used. The other step is iteration. It comprises four substages: subset extraction, model training, prediction-making for missing values, and updating imputation. For each iteration $t = 1, 2, \dots, T$, the subset extraction step includes extraction of independent values as X_{train} and the column with the missing values as the dependent variable, y_{train} , excluding missing y values. Then, these values are fit into a regression model f_j . After that, the values of the predictions of regression model are obtained as $\hat{X}_{ij}^{(t)} = f_j(X_{i,\text{train}})$ which are used rather than missing values at the end. Nevertheless, these predictions are updated until they meet the specified criteria using Equation 1.

$$\hat{X}_{ij}^{(t)} = \begin{cases} X_{ij} & \text{if } M_{ij} = 0 \\ f_j(X_{i,\text{train}}) & \text{if } M_{ij} = 1 \end{cases} \quad (1)$$

where $\hat{X}_{ij}^{(t)}$ is the estimated value of the missing value at i -th row and j -th column at iteration t , X_{ij} is the original value of the data point, M_{ij} is the matrix of missing values (it takes two values 0 for missing values, and 1 for other values), f_j is the used regression model, and $X_{i,\text{train}}$ is the used data while training f_j .

B. DATA NORMALIZATION

Almost all the time, dataset different features differ in their ranges and their units. As a consequence, some features may range on a small scale and others on a huge scale, which negatively affects the models while making predictions. Based on that, the features must be normalized in the same way. One technique that transforms the features to a fixed range is Min Max Scaling technique [44]. It always scales the features between 0 and 1. However, it also can be used to scale data

into other fixed ranges, see Equation 2.

$$X' = a + \frac{(X - X_{\min})(b - a)}{X_{\max} - X_{\min}} \quad (2)$$

where X' is the scaled value, X is the original value, $X_{\min}$ is the minimum value of that feature, $X_{\max}$ is the maximum value of that feature, a is the minimum range, and b is the maximum range. When the required scale is $[0, 1]$, then, the values of a and b must be 0 and 1, respectively.

C. CHANGE POINTS DETECTION

Detecting change points in time series is crucial for improving forecasting accuracy. One effective method for identifying these points is Binary Segmentation. In short, it splits the data repetitively at the point that maximizes a given norm e.g., a cost function, such as MSE, until it is met. Consequently, this method's key working principle rotates around three main points: initialization, recursive segmentation, and stopping norm [45]. Given a time series $\mathbf{y} = \{y_1, y_2, \dots, y_n\}$, the cost function is calculated for the $\mathbf{y}_{a:b}$ segment, where a is the start point and b is the end point of the segment, using Equation 3.

$$C(\mathbf{y}_{a:b}) = \sum_{i=a}^b (y_i - \bar{y}_{a:b})^2 \quad (3)$$

where $\bar{y}_{a:b}$ is the mean. Based on that, the optimal change points τ_1 are found by minimizing the total cost using Equation 4.

$$\tau_1 = \arg \min_{\tau} [C(\mathbf{y}_{1:\tau}) + C(\mathbf{y}_{\tau+1:n})] \quad (4)$$

The process is repetitively applied to the different segments $y_{1:\tau_1}$ and $y_{\tau_1+1:n}$.

D. ARTIFICIAL NEURAL NETWORK

1) RECTIFIED LINEAR UNIT (ReLU)

ReLU is a type of activation function which determines whether a neuron should be activated or not, see Equation 5. Such functions are commonly used in ANNs and deep learning models as they play crucial role in increasing simplicity and non-linearity of the network [46]. They are essential in back-propagation process as well.

$$\text{ReLU}(x) = \max(0, x) \quad (5)$$

2) FULLY CONNECTED LAYER

ANNs, in general, consist of two main components: layers and neurons. A layer is called fully connected when the neuron of one layer is totally connected with the neurons of the other layer so that it gets inputs from them all. After receiving the inputs, it does three essential steps, which are performing the weighted sum of these inputs ($\sum_{j=1}^n w_{ij}x_j$) where x_j is the input, adding bias b to them, and applying an activation function Φ on the result.

3) LSTM

It is an RNN model that is first introduced to the memory cells and gating mechanisms [47]. It mainly capturing information from the series data points using cell state (c_t) and hidden state (h_t). The value of the first and the second one are calculated using Equation 6 and Equation 7, respectively. But for the value of c_t to be calculated, the value of the candidate cell state ($c_{\sim t}$) must be firstly found using Equation 8.

$$c_t = f_t \odot c_{t-1} + i_t \odot c_t \quad (6)$$

$$h_t = o_t \odot g(c_t) \quad (7)$$

$$c_t = \tanh(W_c x_t + U_c h_{t-1} + b_c) \quad (8)$$

In addition, the values of the three different control gates must also be calculated: Equation 9 for the input gate (i_t), Equation 10 for the forget gate (f_t), and Equation 11 for the output gate (o_t).

$$i_t = \sigma(W_i x_t + U_i h_{t-1} + b_i) \quad (9)$$

$$f_t = \sigma(W_f x_t + U_f h_{t-1} + b_f) \quad (10)$$

$$o_t = \sigma(W_o x_t + U_o h_{t-1} + b_o) \quad (11)$$

where σ is the activation function, (usually the sigmoid function), W is the weight matrix, x_t is the input at time step t , U is the weight matrix for the hidden state from the previous time step $t-1$, h_{t-1} is the hidden state from the previous time step, and b is the bias term.

E. HYPERPARAMETER TUNING TECHNIQUES

Grid search is a hyperparameter optimization technique used to find the optimal set of hyperparameters for a machine learning model. Its working principle can be formally explained as follows: let θ represent a set of hyperparameters, and let $\mathcal{M}(\theta)$ be the machine learning model. The Grid search technique finds the optimal hyperparameters that maximize the performance metric on the validation $\mathcal{L}$ data by trying each possible combination and evaluating them on the model itself. However, the main disadvantage of this method is that it is computationally expensive, especially when the number of hyperparameter sets is large. In order to overcome this problem, the random search technique is used. It only performs random selection from the given hyperparameter sets [48].

Nevertheless, such techniques may not be useful to be implemented for huge ANNs. As a result, the trial-and-error approach is used. During it, both the evaluation metrics and losses of the training and validation processes are closely monitored. For instance, if the validation loss is significantly higher than the training loss, it typically indicates that the model is overfitting. Conversely, if the training loss does not improve, the model is likely underfitting. Furthermore, if the model's loss drops abruptly at the beginning, it suggests that the learning rate is too high, whereas a slowly decreasing loss indicates a low learning rate. Based on these observations, hyperparameters are adjusted through various trials. To expedite the process, the early stopping technique can be employed, which halts training if there is no improvement in the specified metric after a certain number of epochs.

F. WEIGHTS AND BIASES INITIALIZATION

One method used to initialize the weights of ANNs is Glorot uniform (also known as Xavier initialization) [49]. It aims to keep the scale of the gradients roughly the same in all layers of the network. The initial weights are initialized within the range $\left[-\sqrt{\frac{6}{n_{in} + n_{out}}}, \sqrt{\frac{6}{n_{in} + n_{out}}}\right]$, where n_{in} is the number of input units in the weight tensor and n_{out} is the number of output units.

G. EVALUATION METRICS

MSE quantifies the average squared difference between predicted and actual values $\left(\text{MSE} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2\right)$. Where n is the number of data points, y_i is the actual value, and $\hat{y}_i$ is the predicted value. On the other hand, RMSE provides the square root of MSE, $\left(\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}\right)$ offering error metrics in the same units as the data. Furthermore, MAE quantifies the average absolute difference between predicted and actual values $\left(\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|\right)$. Finally, R^2 score indicates the proportion of the variance in the dependent variable that is predictable from the independent variables $\left(R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2}\right)$.

V. SYSTEM MAIN ARCHITECTURE

The data pipeline architecture of this study, illustrated in Figure 2, comprises six distinct stages. In the initial stage, data is collected from four primary open data sources using Selenium within Python, and then published to four Apache Kafka topics. Once all data is collected, it is consolidated into a single file containing 19 features, including date-time, independent variables, and dependent variable. The data is completely explained in section VII: DATA OVERVIEW. The second stage involves preprocessing steps, which include data cleaning, transformation, and feature engineering. Data cleaning addresses missing values and irrelevant data, while transformation employs Min-Max scaling. Feature engineering introduces new features, performs multicollinearity checks, and utilizes Principal Component Analysis (PCA) to reduce data dimensionality. Appropriate explanation of each method is provided in Section VIII: EXPERIMENTAL ANALYSIS. The third stage is a conventional data splitting process, where the data is divided into training, validation, and test sets. Both the validation and test sets encompass 84 days, approximately three months of data each, leaving 2113 days for the training data. Up to this point, all processes have focused on data acquisition and preprocessing. Now, the data is ready for model application. At this phase, The previously mentioned 21 built-in forecasting models were developed using scikit-learn and TensorFlow within Python and have been meticulously tuned to optimize performance. These models are evaluated on both the validation and test datasets, covering six months of data in total. Additionally, a novel model, ELSTMD, is constructed from scratch. The

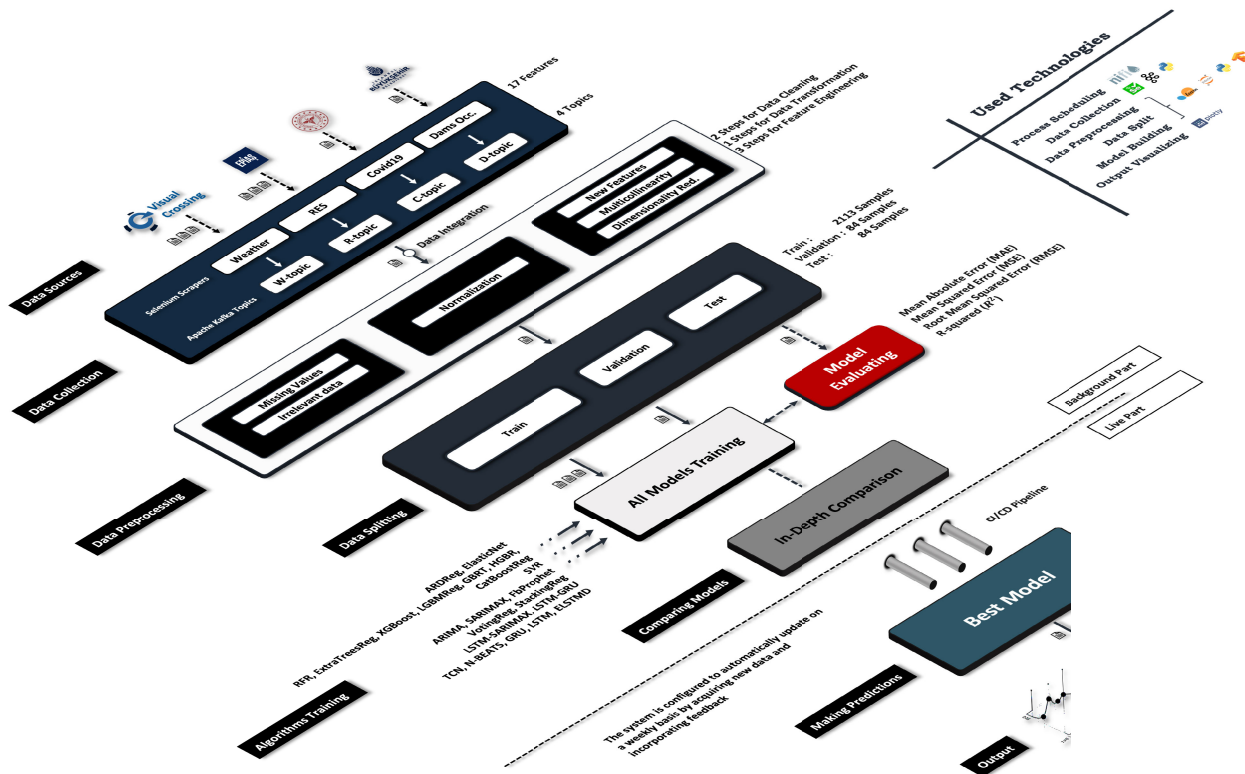


FIGURE 2. Detailed workflow of the principal machine learning pipeline.

predictions of each model are compared using various metrics, including R^2 score, MAE, MSE, and RMSE. Ultimately, the model yielding the best results is selected for implementation in the live system.

VI. ELSTMD MODEL

In this section, we embark on an in-depth exploration of the model, aiming to provide a clear understanding of its formulation, the reliability of its reproduction across various scenarios, and the challenges presented by its current constraints.

A. INTERNAL STRUCTURE

The proposed model (ELSTMD) architecture is inspired by the encoder-decoder model structure [50]. However, instead of a traditional bottleneck layer, it incorporates a high-capacity LSTM layer. The overall architecture comprises nine principal layers; however, Figure 3 displays only seven, as the two reshaping layers employed exclusively during training have been omitted for clarity.

The detailed internal structure of ELSTMD model is presented in Figure 4. The first layer only defines the input shape. The second one reshapes the input to a format suitable for training process which is critical for the model compatibility to understand the dimensions of the data. Specifically, given an input with D features (dimensions) over T timesteps, the input is $\mathbf{x} \in \mathbb{R}^{T \times D}$. This is

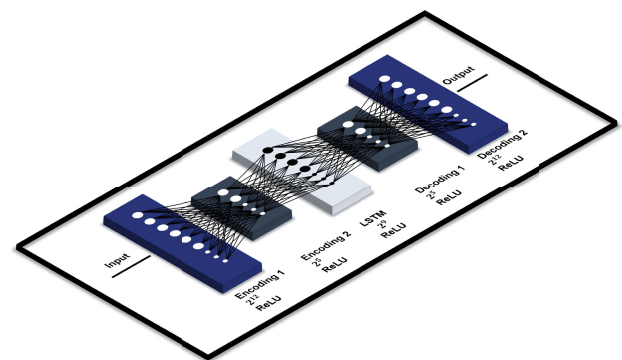


FIGURE 3. Layered architecture of the ELSTMD model.

reshaped to $\mathbf{x}' = \text{reshape}(\mathbf{x}, (T \cdot D, 1)) = \mathbf{x} \in \mathbb{R}^{TD}$. The first encoding layer is a fully connected layer, producing the output $\mathbf{h}_1 = \text{ReLU}(\sum_{j=1}^n W_1 x'_j + \mathbf{b}_1)$ where $\mathbf{W}_1 \in \mathbb{R}^{2^{12} \times TD}$, $\mathbf{b}_1 \in \mathbb{R}^{2^{12}}$. The second encoding layer takes $\mathbf{h}_1$ as input and produces $\mathbf{h}_2 = \text{ReLU}(W_2 \mathbf{h}_1 + \mathbf{b}_2)$ where $W_2 \in \mathbb{R}^{2^5 \times 2^{12}}$, $\mathbf{b}_2 \in \mathbb{R}^{2^5}$. The previous two layers introduce non-linearity and learn abstract features from the input data through linear transformations and activation functions. Next, $\mathbf{h}_2$ is reshaped to $\mathbf{h}'_2 = \text{reshape}(\mathbf{h}_2, (T, 2^5)) = \mathbf{h}_2 \in \mathbb{R}^{T \times 2^5}$ so that the middle LSTM layer can process it. The LSTM layer outputs $\mathbf{h}_3 = \text{LSTM}(\mathbf{h}'_2)$ where

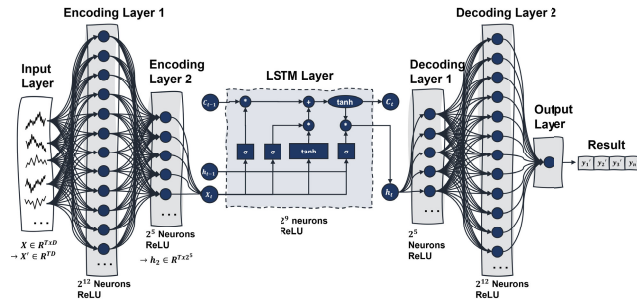


FIGURE 4. The internal structure of ELSTMD model.

$h_3 \in \mathbb{R}^{2^9}$. It is used in order to capture additional temporal dependencies. After that, the first decoding layer takes h_3 as input and produces $h_4 = \text{ReLU}(W_3 h_3 + b_3)$ where $W_3 \in \mathbb{R}^{2^5 \times 2^9}$, $b_3 \in \mathbb{R}^{2^5}$. The second decoding layer takes h_4 as input and produces $h_5 = \text{ReLU}(W_4 h_4 + b_4)$ where $W_4 \in \mathbb{R}^{12 \times 2^5}$, $b_4 \in \mathbb{R}^{12}$. The decoding part applies further transformations and non-linearity. Finally, the output layer produces $y = W_5 h_5 + b_5$ where $W_5 \in \mathbb{R}^{1 \times 2^{12}}$, $b_5 \in \mathbb{R}^1$.

A detailed layer-wise summary of the model, including the number of neurons, output shapes, and parameter counts, is presented in Table 1, below. It is clear that the reshape layers lack trainable parameters since they do not contribute to the training process. The output shapes of the encoding and decoding parts are identical. Additionally, the LSTM layer possesses the highest number of trainable parameters.

B. MODEL REPRODUCIBILITY

This section details the reproducibility aspects of the ELSTMD model, including its experimental setup and hyperparameter configuration. Data sources and the dataset are thoroughly described in Section VII: DATA OVERVIEW, while the data preparation techniques are outlined in Section VIII: EXPERIMENTAL ANALYSIS. The ELSTMD model employs the hyperparameters listed in Table 2 and comprises a total of 1,427,521 trainable parameters. The model was trained over 95 epochs with a batch size of 8 using the Adam optimizer at a learning rate of 3×10^{-3} and activation function of ReLU for all layers. All kernels are initialized using the Glorot uniform method, and all biases are set to zero. To mitigate overfitting, a ridge regression (L2) kernel weight regularizer with a factor of 0.016 is applied to both the second encoding and decoding layers. In the ELSTMD model, the hyperparameter $n_{\text{timesteps}} \in \mathbb{Z}^+$ specifies how many consecutive time steps from the sequence $X \in \mathbb{R}^{N \times d}$, where N is the number of samples and d is the number of features per sample, are used to form each overlapping input window. For example, when $n_{\text{timesteps}} = 5$, each window consists of the samples $(x_t, x_{t+1}, x_{t+2}, x_{t+3}, x_{t+4})$ for $t = 1, \dots, N - 4$, where each $x_i \in \mathbb{R}^d$. This sliding-window approach provides the model with local temporal context in each training example, which is subsequently flattened or reshaped, passed through Dense (encoder) layers, processed by the LSTM to capture temporal dependencies, and finally mapped to the desired output via the decoder Dense layers.

TABLE 1. Layer-wise ELSTMD model's internal structure.

Layer	Neurons #	Output Shape	Param #
Input	0	(None, T, D)	0
Reshape 1	0	(None, D)	0
Encoding 1	2^{12}	(None, 4096)	24576
Encoding 2	2^5	(None, 32)	131104
Reshape 2	0	(None, 1, 32)	0
LSTM	2^9	(None, 512)	1116160
Decoding 1	2^5	(None, 32)	16416
Decoding 2	2^{12}	(None, 4096)	135168
Output	1	(None, 1)	4097

TABLE 2. Overview of ELSTMD model hyperparameters.

Hyperparameter	Value
Enc_1 & Dec_1 neurons	2×10^{12}
LSTM neurons	2×10^5
Enc_2 & Dec_2 neurons	2×10^9
Enc_2 & Dec_2 L2	0.016
$n_{\text{timesteps}}$	10
Activation Function	ReLU
Optimizer	Adam
Learning Rate	3×10^{-3}
Batch Size	2^3
Epochs	95
Weights Initializer	Glorot Uniform
Bias Initializer	Zeros

Choosing a larger $n_{\text{timesteps}}$ allows the model to capture longer-range dependencies at the cost of higher complexity, whereas a smaller value reduces the temporal field of view but can help prevent overfitting. Training was executed on a system equipped with 16 CPU cores and one Nvidia RTX 4070 GPU (8GB RAM). The training environment comprises Python 3.12.9 and TensorFlow 2.18.0.

C. COMPUTATIONAL COST, SCALABILITY, AND LIMITATIONS

The ELSTMD model contains 1,427,521 trainable parameters, which reflects a balanced level of complexity and computational efficiency. This moderate parameter count means that the model can be trained and deployed on standard hardware configurations without demanding excessive computational resources or memory. Consequently, training can be performed on commonly available CPUs and GPUs, enabling quicker experimentation and iteration. From a scalability perspective, the modest size of the model offers several benefits:

- **Efficient Training:** The manageable number of parameters allows for faster convergence during training, reducing both the time and computational cost.
- **Resource Flexibility:** Its efficiency makes it suitable for deployment in resource-constrained environments while still delivering robust performance.
- **Extensibility:** The architecture can be further fine-tuned or expanded as needed, allowing for scalability to larger datasets or more complex tasks without incurring prohibitive computational overhead.
- **Real-Time Forecasting:** Due to its computational efficiency, the ELSTMD model is well-suited for real-time

forecasting applications, delivering timely and accurate predictions in dynamic environments.

The key factors affecting computational efficiency can be summarized as the number of parameters, forward pass complexity, and the depth of the computation graph. The ELSTMD model does require more computation than simpler single-layer or feed-forward approaches. However, these added costs often lead to improvements in accuracy and representational power. The precise trade-off depends on the size of the training dataset, the nature of the task, and the available computational resources.

The model's limitations can be summarized in five main points. First, despite its moderate size (1,427,521 trainable parameters), memory usage may become a bottleneck when scaling to larger datasets or more complex architectures, making efficient memory management crucial for realtime applications. After that, as the model scales, computational costs may increase nonlinearly, requiring additional resources or optimizations. Third, proper regularization is essential to prevent overfitting, although the optimal strategy depends on the specific problem. Fourth, reliance on specific Python and TensorFlow versions can lead to compatibility issues that affect reproducibility and deployment. Eventually, hyperparameters must be selected properly; otherwise, issues such as underfitting or overfitting can occur, further impacting the model's effectiveness.

VII. DATA OVERVIEW

Incorporating extensive historical time-series datasets generally enhances the accuracy of AI models. However, increasing the number of features to a high number may reduce accuracy as the data becomes distributed across multiple dimensions, complicating the process and raising computational costs. As a result, the dataset used is on a daily basis, ranging from January 1, 2018, to March 31, 2024, comprising 2,281 rows. As stated in Table 3, the data includes 17 independent variables, 1 date-time variable, and 1 dependent variable. The independent variables can be categorized into five main categories. The electricity-related category consists of electricity generated from fuel oil and natural gas in MWd, and maximum RES capacity in MWd. In addition, the weather category includes cloud cover percentage of the sky, humidity in (g/kg), rain precipitation in mm, pressure in (Pa), snow in inches, solar radiation in (kWh/m²), temperature in (°C), and wind speed in (m/s). Moreover, the economic-related features are RES unit cost in (TL/MWd) and GDP in (1000\$/year). As a natural disaster, only COVID19 cases are considered. Additionally, there are two self-generated features: season and change points, which are related to the summer and winter seasons, and abrupt changes in the RES generation, respectively. If the condition is satisfied, then their value is one; if not, it is 0. It is important to note that all the previously mentioned features are related to Turkey.

In addition, an open-source dataset, as referenced in [58], is employed to evaluate the generalization capabilities of the

TABLE 3. Dataset features and corresponding information.

Feature	Mean	Std.	Min	Max	Ref.
RES G. (y)	170964.9	52402.5	54295.2	349413.2	[42]
Fuel Oil (X_1)	2012.0	1297.4	303.9	9031.8	[51]
Cloud (X_2)	44.1	28.4	0.0	99.6	[52]
Covid19 (X_3)	5676.2	14647.5	0.0	111157.0	[53]
Dam Occ. (X_4)	55.1	21.5	13.3	95.5	[54]
GDP (X_5)	4.5	3.4	0.8	11.4	[55]
Humidity (X_6)	57.9	16.7	14.0	97.0	[52]
RES C. (X_7)	28041.6	3103.7	22034.4	33369.5	[56]
Natural G. (X_8)	206430.9	89556.6	17235.4	443433.8	[51]
Precipi. (X_9)	1.2	4.3	0.0	86.9	[52]
Pressure (X_{10})	1014.6	6.4	994.8	1034.7	[52]
RES U.C. (X_{11})	1049.3	1072.0	160.0	3883.9	[57]
Snow (X_{12})	0.1	0.6	0.0	10.8	[52]
Solar R. (X_{13})	200.3	89.1	5.9	369.9	[52]
Temp. (X_{14})	13.3	8.5	-8.8	33.4	[52]
Wind S. (X_{15})	49.1	20.6	9.0	154.8	[52]
Season (X_{16})	0.5	0.5	0.0	1.0	P.
Cha. Ps (X_{17})	0.1	0.2	0.0	1.0	P.

ELSTMD model over short time horizons. The dataset was collected from two solar power plants in India during a 34-day period and comprises both inverter-level power generation data and plant-level sensor readings. The power generation dataset, recorded at 15-minute intervals, originally included DC and AC power outputs along with daily and cumulative inverter yields. However, due to the high correlation among these metrics, only the cumulative inverter yield—serving as the target (dependent feature)—was retained. Similarly, while the sensor data provides ambient and module temperature measurements as well as irradiation levels, only ambient temperature is preserved. The dataset spans from May 15, 2020 00:00:00 to June 17, 2020 23:45:00, covering the entire 34-day period. The season and change points are generated for this dataset as well. Nevertheless, the season is determined based on the highest-temperature hours, setting it to 1 for hours between 11:00 and 17:00 and to 0 for the remaining hours, given that the data range is limited.

The model is evaluated using datasets sourced from both Turkey and India, each reflecting unique seasonal characteristics. For season feature, the Turkish dataset captured two distinct seasons—summer and winter—reflecting clear climatic patterns. In contrast, India's dataset, though constrained by its smaller size, captured key variations by focusing on sun hours. These assumptions facilitated an initial demonstration of the model's effectiveness under varied regional conditions. However, future research should expand to additional geographical contexts and more granular seasonal definitions to comprehensively assess model generalization.

VIII. EXPERIMENTAL ANALYSIS

In this section, we provide a comprehensive overview of the implementation process and present experimental results for both the ELSTMD model and the comparison models. Specifically, the implementation details and analysis are thoroughly explained for the first Turkey dataset, whereas for the second India dataset, only the experimental results are presented.

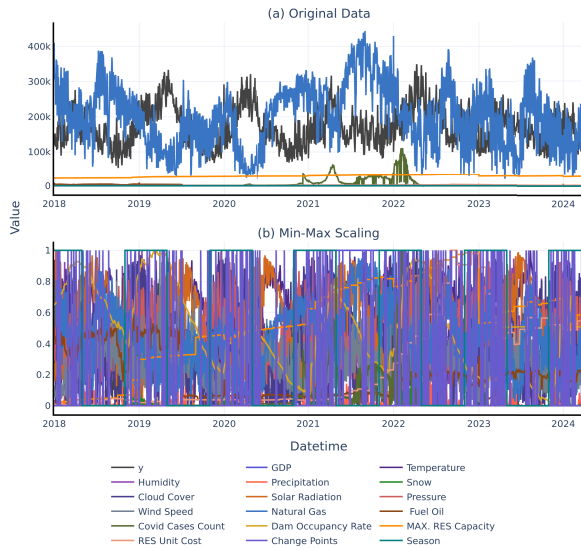


FIGURE 5. Raw data comparison: (a) Original vs. (b) Min-Max scaled.

A. IMPLEMENTATION DETAILS

As mentioned earlier, the first data are collected from four distinct open sources. A data harmonization process is then applied to align all data to a uniform row size and merge them into a single file. In total, fifteen distinct features are collected, and two additional features are generated. Furthermore, a second dataset—related to two solar power plants in India—is employed to evaluate the generalization of the ELSTMD model across different datasets. For more details regarding the datasets, please refer to Section VII: DATA OVERVIEW. All subsequent operations are thoroughly explained using the first dataset (related to Turkey), which is used for long horizon forecasting. The same procedures are applied to the second dataset (related to India), which is designated for shorter horizon, specifically one day ahead, forecasting. These operations encompass three main steps: data preprocessing (including imputation and scaling), feature engineering (comprising change point detection and seasonality analysis), and feature orthogonalization (incorporating multicollinearity analysis and dimensionality reduction).

1) DATA PREPROCESSING: IMPUTATION AND SCALING

However, while investigating the data, it is found that the dataset includes two features, dam occupancy rate, and solar radiation, which have 171 and 90 missing values, respectively. Based on that, the first applied process is imputing them. These missing values were imputed using the iterative imputation approach with a Bayesian Ridge model over 20 iterations. Subsequently, an irrelevant value was identified in the Covid-19 feature, specifically on December 12, 2020. To address this issue, the anomalous value was replaced with the mean value calculated from the same data points. These two steps ensure the integrity and increase the accuracy of the dataset. The obtained raw data, along with the labels of each feature, are completely displayed in Figure 5 (a). It is evident



FIGURE 6. RESs power generation with detected Change points.

from this figure that each feature varies in its range, as they possess different units and represent distinct quantities. This diversity in measurement scales underscores the necessity for normalization or standardization. In order to overcome this problem, each feature is scaled using Min Max Scaler based on $[0, 1]$ range, see Figure 5 (b).

2) FEATURE ENGINEERING: CHANGE POINTS AND SEASONALITY

As part of feature engineering, two new features, ‘season’ and ‘change points’, were created. The ‘season’ feature is based on the months of winter and summer in Turkey, with winter months assigned a value of 1 and summer months a value of 0. The ‘change points’ feature is determined using the Binary Segmentation approach, identifying 150 change points, as clearly illustrated in Figure 6. These change points are assigned a value of 1, while all other points are given a value of 0.

3) FEATURE ORTHOGONALIZATION: MULTICOLLINEARITY ANALYSIS AND DIMENSIONALITY REDUCTION

As it is well known that features with multicollinearity reduce the generalization ability of models, a multicollinearity check process was implemented. While other methods exist for the same purpose—such as feature selection using various models—this study employed the Variance Inflation Factor (VIF) to assess the collinearity among independent features. VIF is a commonly used technique for detecting multicollinearity, and it can be calculated using Equation 12.

$$\text{VIF}(X_i) = \frac{1}{1 - R_i^2}, \quad (12)$$

where X_i is the i -th independent variable in the regression model, and R^2 is the coefficient of determination obtained by regressing X_i on all other predictors. The threshold of VIF for determining collinearity among independent variables is not fixed and can depend on the specific problem. However, practitioners often rely too heavily on fixed VIF or tolerance thresholds (e.g., 4 or 10) as rules of thumb for indicating excessive multicollinearity [59], [60].

Consequently, we stated that if any feature has a VIF equal to or more than 9, then the feature has a collinearity phenomenon and is dropped. As a result, 9 out of the 17 features are dropped: temperature, solar radiation, maximum RES capacity, humidity, dam occupancy rate, pressure,

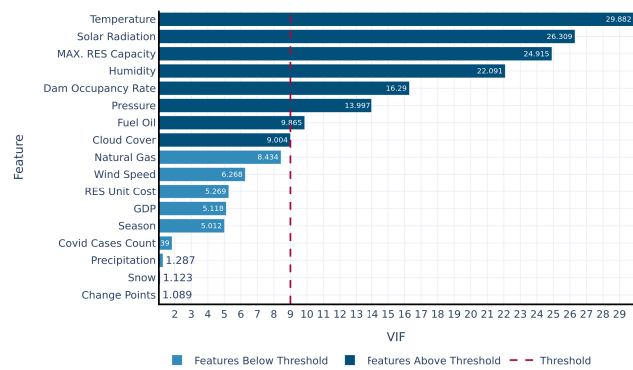


FIGURE 7. Feature multicollinearity analysis.

fuel oil, and cloud cover. It is obvious that these features somehow show a similar trend with the other remaining features, so they must be dropped. This step is crucial as it increases the robustness of regression models. The value of VIF for each feature is presented in Figure 7. Features with Cerulean are considered as features without multicollinearity phenomenon, features with dark Cerulean are features with collinearity phenomenon as they are passing the threshold, and the vertical dashed line with rich Carmine color.

Moreover, PCA was employed to further reduce the dimensionality of the dataset, with the goal of preserving 90% of the total variance. This threshold is widely adopted in practice because it balances reducing computational complexity against retaining the majority of information present in the original data. A variance cutoff in the range of 70–90% is typically sensible for PCA, as it preserves the essential structure of the data while achieving meaningful dimensionality reduction. However, the optimal threshold may vary based on dataset-specific considerations [61], [62]. Based on our experiments, capturing 90% of the data’s variance was found sufficient to represent the key underlying patterns over time without excessively sacrificing interpretability or computational efficiency. As a result, when PCA was applied to the data, it identified 5 principal components, reducing the number of input independent features from 9 to 5 PCs. This approach becomes especially vital for larger datasets because it minimizes computational cost without negatively affecting the model’s performance. The individual explained variance and the cumulative explained variance are displayed in Figure 8 as bar and line charts, respectively.

By applying VIF and PCA, the raw data is changed from the one shown in Figure 5 (b) to the one in Figure 9 which shows the remaining dependent and PC features as separated line charts. It is clear that the complexity of the input data has been significantly reduced by selecting only the most relevant features. This approach enhances the robustness and validity of the models while simultaneously lowering computational costs. By eliminating similar and redundant features, the tendency of models to memorize rather than generalize is also diminished. This careful feature selection process ensures that the models focus on the most impactful data, leading to more reliable and efficient performance.

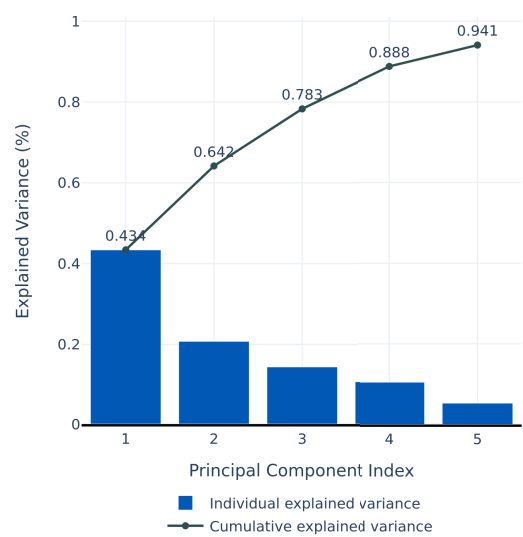


FIGURE 8. PC explained variance: Individual and Cumulative.

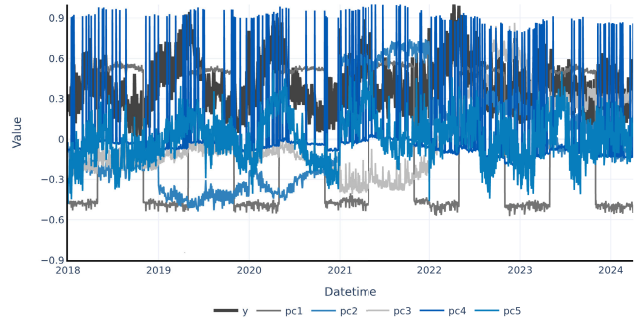


FIGURE 9. PCs and Y-Values: Final model data.

B. RESULTS AND DISCUSSION

The results and discussion part presents a rigorous numerical analysis paired with a thorough evaluation of model interpretability, featuring insights into feature importance and an ablation study that highlight key performance dynamics.

1) NUMERICAL ANALYSIS

The 21 models were prepared and evaluated using various metrics such as R^2 score, MSE, MAE, and RMSE. These models were tuned to achieve optimal performance. In general, the best hyperparameters of each model are found using Grid search technique. Moreover, Random search technique is also used for the same purpose. However, Grid search technique cannot be used for some models, therefore, trial-and-error method next to monitoring the training and validation process are used to determine the best hyperparameters. In addition to these models, we developed a custom model from scratch intending to surpass their performance. The proposed model demonstrated superior performance compared to all previously mentioned models, as detailed in Table 4, where T. T. represents the training process time. As the R^2 metric is positive for all models, it can be concluded that all models perform better than using the mean as a predictive benchmark. The previously mentioned models are all trained using 16 CPU cores and GPU of Nvidia RTX

TABLE 4. Model-based performance metrics comparison.

Model	R ²	MAE	MSE	RMSE	T. T.
ElasticNet	0.00203	0.10672	0.01593	0.12623	0.0001
ExtraTreesReg	0.06558	0.10293	0.01492	0.12215	0.2684
RFR	0.10123	0.09949	0.01435	0.11979	0.311
StackingReg	0.14592	0.09061	0.01364	0.11678	1.1929
N-BEATS	0.15976	0.0972	0.01342	0.11583	0.8271
ARDReg	0.16551	0.09684	0.01332	0.11543	0.0001
GRU	0.21	0.0922	0.01261	0.11231	0.5255
VotingReg	0.31076	0.08821	0.011	0.1049	0.5771
ARIMA	0.33261	0.08062	0.01066	0.10323	0.1053
TCN	0.40077	0.08256	0.00957	0.09781	6.0019
LSTM-GRU	0.41516	0.08163	0.00934	0.09663	2.05
XGBoost	0.41581	0.0729	0.00933	0.09658	0.0068
LSTM-SARIMAX	0.43157	0.07934	0.00908	0.09527	1.14
SARIMAX	0.43172	0.07931	0.00907	0.09526	0.1275
SVR	0.46417	0.07038	0.00856	0.0925	0.0003
LGBMReg	0.48562	0.07191	0.00821	0.09062	0.0081
GBRT	0.48886	0.06734	0.00816	0.09034	0.0051
HGBR	0.49781	0.0724	0.00802	0.08955	0.0026
CatBoostReg	0.50537	0.06905	0.0079	0.08887	0.0026
FbProphet	0.52925	0.07294	0.00752	0.0867	0.4179
LSTM	0.54221	0.06879	0.00731	0.08549	1.898
ELSTMD	0.56714	0.06398	0.00691	0.08313	3.4919

4070 8GB RAM. It is evident that linear models exhibit the lowest performance, which is expected given the non-linear nature of the forecasting task. The models yielding the best results belong to the ANN category, affirming that ANNs are well-suited for non-linear forecasting problems. The proposed model achieved R², MAE, MSE, and RMSE values of 0.56714, 0.06398, 0.00691, and 0.08313, respectively. These metrics indicate that our model outperforms all other models, with the LSTM model demonstrating the second-best performance.

To examine the predicted values of the top 10 models compared to the actual values over 10 days, please refer to Table 5. In addition, Figure 10 illustrates the performance of each forecasting model using separate line charts. In these charts, the actual observed data is depicted in red, while the predictions generated by each model are shown in blue. This visual representation allows for a clear comparison between the actual data and the predicted values, highlighting the accuracy and reliability of each model’s forecast. The detailed analysis provided by these charts is somehow essential for identifying the most effective model.

2) MODEL INTERPRETABILITY

The following analysis entirely examines feature importance and an ablation study to elucidate their contributions to overall model performance.

- **Feature Importance** The radial treemap or sunburst chart presented in Figure 11 illustrates how the five principal components (PCs) contribute to each of the five layers (Encoding 1, LSTM, Encoding 2, Decoding 1, and Decoding 2) in the ELSTMD model. The outer ring highlights the relative feature importance of each PC within a specific layer, indicating which components most strongly influence that layer’s transformations. Higher importance scores suggest that a given PC carries

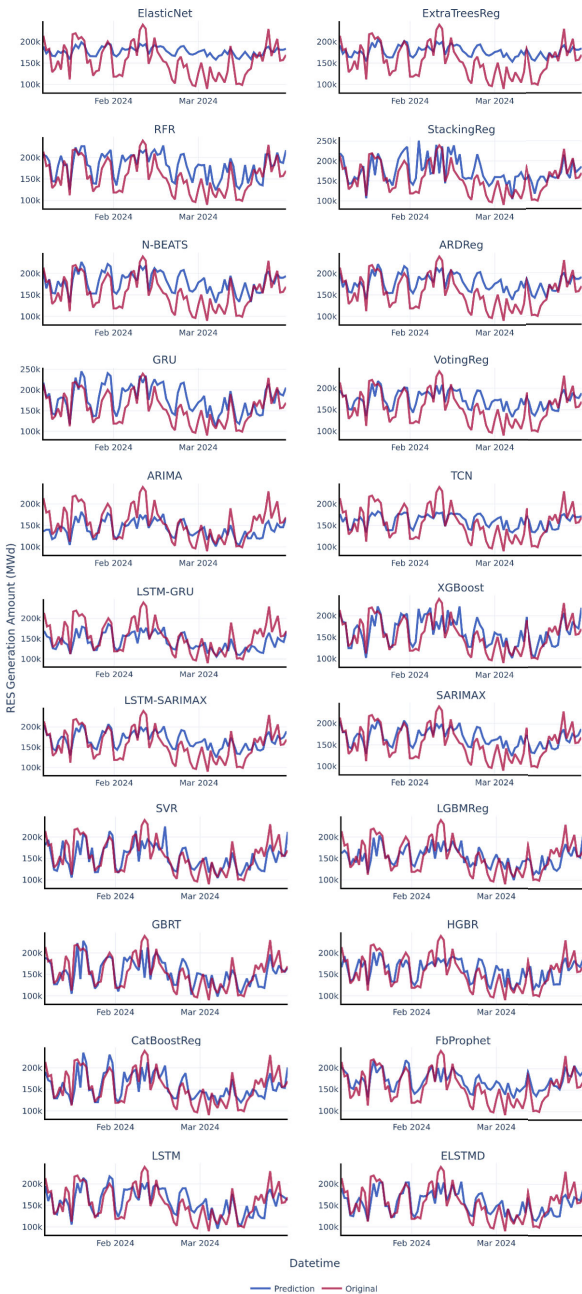


FIGURE 10. Performance evaluation of actual vs predicted for each model.

crucial information for capturing temporal or structural patterns at that stage of processing. Notably, the LSTM layer often emphasizes components associated with dynamic dependencies, reflecting the model’s focus on temporal feature extraction. The encoding and decoding layers show varying degrees of reliance on each PC, underscoring the layered approach of refining and reconstructing the feature representations. Overall, the distribution of feature importance across layers confirms that each principal component plays a distinct yet

TABLE 5. Assessment of top 10 models predictions against actual renewable energy generation (MWD).

Date	Actual Y	ELSTMD	LSTM	FbProphet	CatBoost	HGBR	GBRT	LGBM	SVR	SARIMAX	XGBoost	TCN
2024-01-14	134920.35	166606.41	153283.21	163200.61	166082.16	185232.71	156299.43	172267.05	191718.88	175368.44	177355.94	161832.86
2024-01-15	192856.89	156980.65	165536.53	177729.07	144214.46	167069.4	159471.43	143272.61	145699.21	174465.72	167921.14	168693.93
2024-01-16	179972.39	140971.82	149109.35	166476.92	137626.02	181903.81	147652.77	162552.76	136527.47	157098.44	144162.57	159317.8
2024-01-17	112119.69	114070.12	105363.8	131282.47	114455.69	125745.24	104842.52	116458.05	105518.02	128971.33	101865.83	133779.73
2024-01-18	217002.76	157077.34	165419.75	185964.69	148037.15	156817.11	153197.41	145559.81	145293.37	170243.64	150399.32	168926.54
2024-01-19	219416.70	201158.26	201662.4	216358.07	212576.87	201248.84	222722.25	198584.8	191537.64	196889.05	201200.79	179026.54
2024-01-20	206146.88	169800.36	178641.52	188401.28	154011.21	170094.43	138829.83	155237.4	158058.99	186645.94	175797.43	173213.69
2024-01-21	210551.16	203275.67	213868.31	198049.29	235182.43	200668.72	228106.32	203381.89	207871.7	207497.0	221851.04	182830.43
2024-01-22	202316.97	203123.83	206795.35	205399.22	212576.87	191208.34	214824.28	189186.97	197886.23	193031.09	204970.64	178841.01
2024-01-23	149309.28	148632.25	157116.25	169471.59	147902.12	166024.51	157656.11	153406.63	141865.22	161056.83	162785.13	160319.13

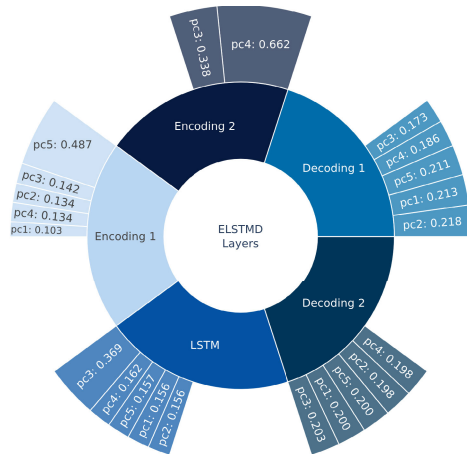


FIGURE 11. ELSTMD model layer-wise analysis of feature importance.

complementary role in driving the model’s predictive performance.

- **Ablation Study** To understand the contribution of each layer in the ELSTMD model, we conducted an ablation study comparing the baseline model with three experimental variants: Variant 1 (excluding the Encoding 1 and Decoding 1 layers), Variant 2 (excluding the Encoding 2 and Decoding 2 layers), and Variant 3 (replacing the LSTM layer with a bottleneck layer). The evolution of training and validation MSE across epochs for these cases is shown separately in Figure 12. In both Variant 1 and Variant 2, the model still learns but ends with a slightly higher loss. “Moreover, in Variant 2, since regularization is excluded, the model starts to overfit the training data as the training MSE decreases, while the validation MSE increases at the 35th epoch. This indicates that each pair of encoding/decoding layers provides unique feature transformations that the network relies on for optimal performance. Variant 3 initially produces a pattern similar to Variant 2. An LSTM layer shares weights across time steps and uses gating mechanisms to selectively retain important information, which limits free parameters, filters out noise, and reduces the risk of memorizing training data. In contrast, a FC layer lacks these benefits, making it more prone to overfitting. Accordingly, when the LSTM layer was replaced with a bottleneck layer, the model began memorizing the training data starting from the 59th epoch, as shown in Variant 3 of Figure 12. Eventually, the baseline model

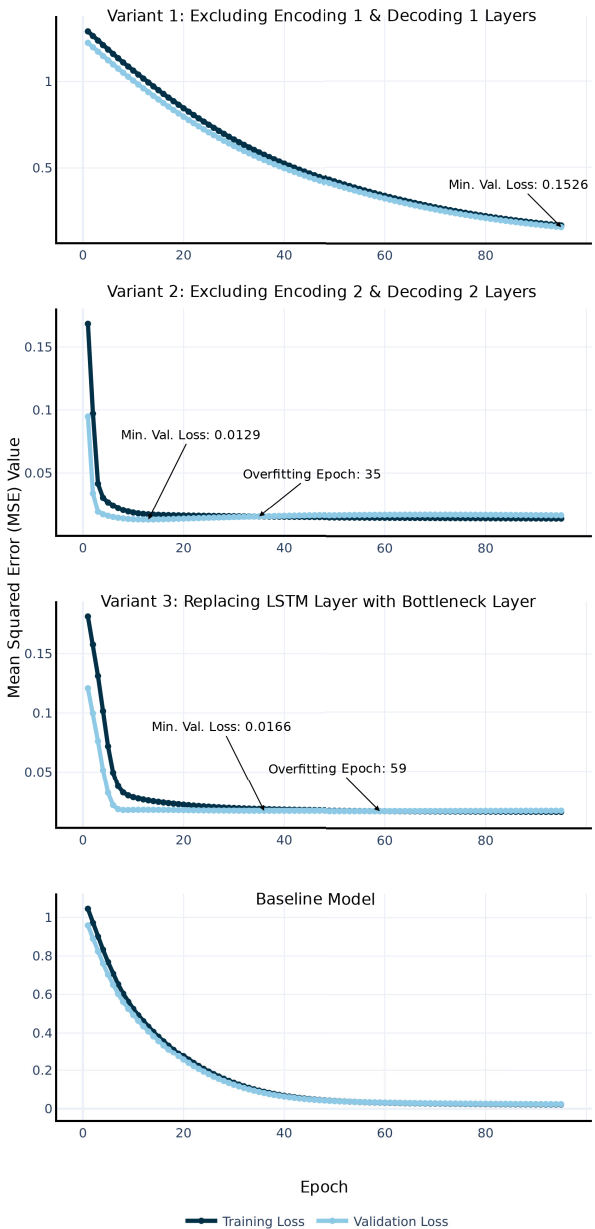


FIGURE 12. Ablation study: Training and validation MSE across epochs.

consistently achieves the lowest MSE at the end of training. With the full complement of layers, the baseline model shows smooth convergence, as both training

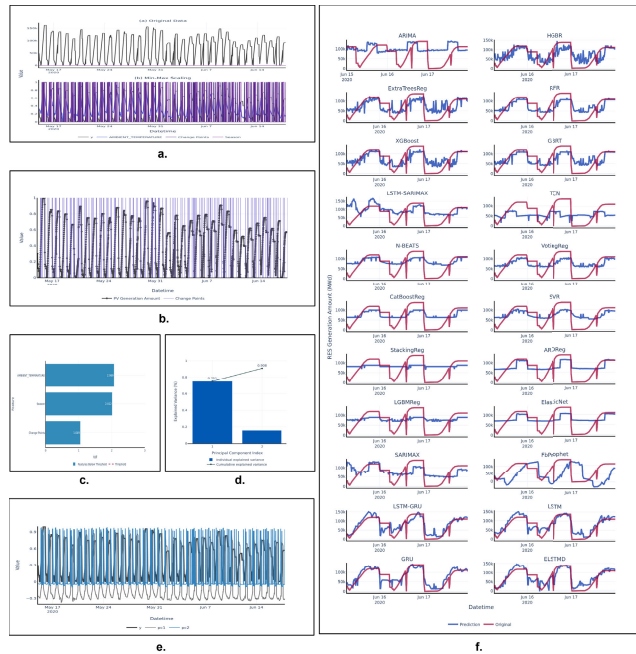


FIGURE 13. India Dataset Results: (a) Original vs. scaled data, (b) Change-point detection and corresponding Y-values, (c) VIF-based multicollinearity analysis, (d) Principal Components explained variance (individual and cumulative), (e) PCs and Y-values for the final model, and (f) Performance evaluation comparing actual vs. predicted values for each model.

and validation losses decrease in tandem, demonstrating that every layer is crucial for robust and effective learning. In summary, each layer in the ELSTMD architecture provides distinct feature transformations, allowing the model to encode both low-level and progressively refined representations of the input. When the first or second encoding-decoding pair is removed (as in Variants 1 and 2), the network loses either the initial feature extraction capability or the subsequent refinement stage, causing it to rely on a shallower transformation pipeline. This reduction in representational depth can lead to higher final loss and earlier onset of overfitting. Similarly, replacing the LSTM with a bottleneck layer compromises the temporal gating mechanisms crucial for selectively retaining and discarding information. Hence, the model becomes more prone to memorizing the training data, as observed in Variant 3. These findings highlight that each component—both encoding decoding pairs and the LSTM layer—plays a vital role in capturing and generalizing complex patterns within the data, ultimately improving overall performance.

Furthermore, the procedures detailed in VIII. EXPERIMENTAL ANALYSIS (specifically A. IMPLEMENTATION DETAILS and B. RESULTS AND DISCUSSION) were also applied to the second dataset related to Indian PV plants, refer to Figure 13. It illustrates six distinct operations—represented across six subfigures. Specifically, subfigure (a) displays the original raw data, and the corresponding min-max scaled version, together constituting the raw data comparison. Sub-

figure (b) depicts PV generation with detected change points, and subfigure (c) presents the feature multicollinearity analysis. Subfigure (d) details the principal component (PC) explained variance analysis (both individual and cumulative), and subfigure (e) displays the most recently obtained data, which serves as the input to the model. Finally, subfigure (f) summarizes the final model data (PCs and Y-values) alongside a performance evaluation comparing actual and predicted values. The ELSTMD model retains the previously discussed structure and hyperparameters, with the exception of three modifications: the learning rate is adjusted to 0.00035, L2 regularization is set to 0.0005, and the number of timesteps is updated to 25. As evident from Figure 13. f., the proposed ELSTMD model, much like its performance on the first dataset, emerges as the leading model among the 21 compared models on the second dataset. It achieves impressive evaluation metrics, including an R^2 score of 0.76574, MAE of 0.09344, MSE of 0.01417, and RMSE of 0.11903.

IX. CONCLUSION

In conclusion, this study comprehensively investigates the effectiveness of various forecasting models for RESs generation, emphasizing the importance of multicollinearity checks and dimensionality reduction techniques to optimize accuracy and computational efficiency. A comparative analysis was conducted across 21 advanced models spanning linear, tree-based, ensemble, ANN, traditional time series, kernel based, and hybrid approaches. To address the challenges in RES forecasting, a novel model, ELSTMD, inspired by the encoder-decoder architecture, was proposed and rigorously tested.

The ELSTMD model was evaluated on two datasets: one capturing long-horizon RES generation in Turkey and the other focusing on short-horizon PV power generation in India. Extensive experimental analysis, including data pre-processing (imputation, scaling), feature engineering (seasonality, change points), feature orthogonalization (multi-collinearity analysis using VIF, dimensionality reduction using PCA), and numerical comparisons (predicted vs. actual values), confirmed the robustness of the model. The study shows that retaining 90% of the data's variance through PCA had no adverse impact on the model's forecasting performance, indicating that this dimensionality reduction technique can be effectively applied to larger datasets without sacrificing predictive accuracy. Furthermore, it underscores the utilization of binary segmentation for detecting change points—a methodology that plays a pivotal role in enhancing the model's forecasting capabilities.

The model achieved superior results compared to the 21 baseline models across both datasets, demonstrating its effectiveness and generalization capability. For the Turkish RES dataset, ELSTMD achieved R^2 of 0.56714, MAE of 0.06398, MSE of 0.00691, and RMSE of 0.08313, while for the Indian PV dataset, it achieved significantly higher accuracy with R^2 of 0.76574, MAE of 0.09344, MSE of 0.01417, and RMSE of 0.11903. These results validate the model's

ability to generalize across different geographical regions and forecasting horizons. Additionally, an extensive layer-wise feature importance and ablation study demonstrated the critical contributions of each model component, ensuring robustness and minimizing overfitting risks.

The ELSTMD model consists of nine layers—one input, two reshaping, and six trainable layers—and encompasses a total of 1,427,521 trainable parameters, striking a balance between complexity and computational efficiency. This moderate size enables efficient training on standard hardware (CPUs and GPUs) while maintaining scalability and adaptability to larger datasets. Key advantages include fast convergence, flexible deployment in resource-constrained environments, and suitability for real-time forecasting applications. However, limitations such as memory constraints on larger datasets, potential increases in computational cost with model scaling, and dependency on specific software versions for reproducibility must be considered. Proper hyperparameter tuning remains essential to mitigate underfitting and overfitting risks.

Future research could further enhance the ELSTMD model by optimizing its architecture, refining hyperparameters, and extending its application to broader forecasting horizons and larger datasets. By incorporating additional features and fine-tuning the neuron count, the model's predictive capabilities could be further strengthened, making it a robust solution for dynamic RES forecasting challenges.

The datasets and the source code relevant to this research are all available on GitHub: <https://github.com/yousefalkhanafseh/ELSTMD>.

REFERENCES

- [1] B. K. Das, M. Hasan, and F. Rashid, "Optimal sizing of a grid-independent PV/diesel/pump-hydro hybrid system: A case study in Bangladesh," *Sustain. Energy Technol. Assessments*, vol. 44, Apr. 2021, Art. no. 100997.
- [2] J. Anable, C. Brand, N. Eyre, R. Layberry, N. Bergman, N. Strachan, and M. Tran, "Lifestyle and energy consumption," UKERC, London, U.K., Tech. Rep. UKERC/WP/ED/2011/001, 2011.
- [3] N. Nyre, J. Anable, C. Brand, R. Layberry, and N. Strachan, "The way we live from now on: Lifestyle and energy consumption," in *Energy*. Evanston, IL, USA: Routledge, 2012, pp. 258–293.
- [4] D. Raimi, Y. Zhu, R. G. Newell, B. C. Prest, and A. Bergman, "Global energy outlook 2023: Sowing the seeds of an energy transition," *Resour. Future*, vol. 1, no. 1, pp. 1–44, 2023.
- [5] Q. Li, G. Xie, and P. Zou, "Energy source and energy dissipation mechanism of coal and gas outburst based on microscope," *Acta Microscopica*, vol. 29, no. 4, p. 1900, 2020.
- [6] D. Balsalobre-Lorente, M. Shahbaz, D. Roubaud, and S. Farhani, "How economic growth, renewable electricity and natural resources contribute to CO₂ emissions?" *Energy Policy*, vol. 113, pp. 356–367, Feb. 2018.
- [7] A. Kona, P. Bertoldi, F. Monforti-Ferrario, S. Rivas, and J. F. Dallemand, "Covenant of mayors signatories leading the way towards 1.5 degree global warming pathway," *Sustain. Cities Soc.*, vol. 41, pp. 568–575, Aug. 2018.
- [8] G. E. Halkos and E.-C. Gkampoura, "Reviewing usage, potentials, and limitations of renewable energy sources," *Energies*, vol. 13, no. 11, p. 2906, Jun. 2020.
- [9] M. E. Hoque, F. Rashid, M. Prodhon, and A. Arman, "Analysis of energy consumption and efficiency to reduce power losses in industrial equipment," in *Proc. Int. Conf. Mech. Eng.*, Dhaka, Bangladesh, Dec. 2017, pp. 18–20.
- [10] O. Ellabban, H. Abu-Rub, and F. Blaabjerg, "Renewable energy resources: Current status, future prospects and their enabling technology," *Renew. Sustain. Energy Rev.*, vol. 39, pp. 748–764, Nov. 2014.
- [11] P. A. Owusu and S. Asumadu-Sarkodie, "A review of renewable energy sources, sustainability issues and climate change mitigation," *Cogent Eng.*, vol. 3, no. 1, Dec. 2016, Art. no. 1167990.
- [12] P. Moriarty and D. Honnery, "The limits of renewable energy," in *Switching Off: Meeting Our Energy Needs in A Constrained Future*. Singapore: Springer, Mar. 2022, pp. 32–54.
- [13] H. Wang, Z. Lei, X. Zhang, B. Zhou, and J. Peng, "A review of deep learning for renewable energy forecasting," *Energy Convers. Manage.*, vol. 198, Oct. 2019, Art. no. 111799.
- [14] L. Frías-Paredes, F. Mallor, M. Gastón-Romeo, and T. León, "Assessing energy forecasting inaccuracy by simultaneously considering temporal and absolute errors," *Energy Convers. Manage.*, vol. 142, pp. 533–546, Jun. 2017.
- [15] B.-M. Hodge, C. B. Martinez-Anido, Q. Wang, E. Chartan, A. Florita, and J. Kiviluoma, "The combined value of wind and solar power forecasting improvements and electricity storage," *Appl. Energy*, vol. 214, pp. 1–15, Mar. 2018.
- [16] L. E. Parker, "Creation of the national artificial intelligence research and development strategic plan," *AI Mag.*, vol. 39, no. 2, pp. 25–32, Jun. 2018.
- [17] L. Yang and M. Zhu, "Review on the status and development trend of AI industry," in *Proc. IEEE 4th Int. Conf. Cloud Comput. Big Data Anal. (ICCCBDA)*, Apr. 2019, pp. 89–93.
- [18] Y. Alkhanafseh, T. C. Akinci, E. Ayaz, and A. A. Martinez-Morales, "Advanced dual RNN architecture for electrical motor fault classification," *IEEE Access*, vol. 12, pp. 2965–2976, 2024.
- [19] M. Khaleel, A. A. Ahmed, and A. Alsharif, "Artificial intelligence in engineering," *Brilliance, Res. Artif. Intell.*, vol. 3, no. 1, pp. 32–42, 2023.
- [20] X. Zhang, Y. Hu, J. Deng, H. Xu, and H. Wen, "Feature engineering and artificial intelligence-supported approaches used for electric powertrain fault diagnosis: A review," *IEEE Access*, vol. 10, pp. 29069–29088, 2022.
- [21] W. Lang, Y. Hu, C. Gong, X. Zhang, H. Xu, and J. Deng, "Artificial intelligence-based technique for fault detection and diagnosis of EV motors: A review," *IEEE Trans. Transport. Electrification*, vol. 8, no. 1, pp. 384–406, Mar. 2022.
- [22] T. Ahmad, H. Chen, R. Huang, G. Yabin, J. Wang, J. Shair, H. M. A. Akram, S. A. H. Mohsan, and M. Kazim, "Supervised based machine learning models for short, medium and long-term energy prediction in distinct building environment," *Energy*, vol. 158, pp. 17–32, Sep. 2018.
- [23] S. Makridakis, E. Spiliotis, and V. Assimakopoulos, "Statistical and Machine Learning forecasting methods: Concerns and ways forward," *PLoS One*, vol. 13, no. 3, 2018, Art. no. e0194889.
- [24] S. S. Soman, H. Zareipour, O. Malik, and P. Mandal, "A review of wind power and wind speed forecasting methods with different time horizons," in *Proc. North Amer. Power Symp.*, Sep. 2010, pp. 1–8.
- [25] H. M. Ertugrul, M. T. Kartal, S. K. Depren, and U. Soytas, "Determinants of electricity prices in turkey: An application of machine learning and time series models," *Energies*, vol. 15, no. 20, p. 7512, Oct. 2022.
- [26] Republic of Turkey Ministry of Energy and Natural Resources. *Enerji ve Madencilik Sektörüne 181,7 Milyar Lira Kamu Yatırımı Yapacağız*. enerji.gov.tr. Accessed: May 12, 2024 [Online]. Available: <https://enerji.gov.tr/haber-detay?id=21231>
- [27] B. Labudović and F. Barbir, "Obnovljivi izvori energije," in *Energetika Marketing*. Rijeka, Croatia: Zagreb, 2002, pp. 1–832.
- [28] B. Abraham and J. Ledolter, *Statistical Methods for Forecasting*. Hoboken, NJ, USA: Wiley, 2009.
- [29] W.-Y. Chang, "A literature review of wind forecasting methods," *J. Power Energy Eng.*, vol. 2, no. 4, pp. 161–168, 2014.
- [30] X. Wang, P. Guo, and X. Huang, "A review of wind power forecasting models," *Energy Proc.*, vol. 12, pp. 770–778, Jan. 2011.
- [31] Y. Hao and C. Tian, "A novel two-stage forecasting model based on error factor and ensemble method for multi-step wind power forecasting," *Appl. Energy*, vol. 238, pp. 368–383, Mar. 2019.
- [32] C. Chatfield, *Time-Series Forecasting*. London, U.K.: Chapman & Hall, 2000.
- [33] V. Assimakopoulos and K. Nikolopoulos, "The theta model: A decomposition approach to forecasting," *Int. J. Forecasting*, vol. 16, no. 4, pp. 521–530, Oct. 2000.
- [34] V. Cerqueira, L. Torgo, and C. Soares, "Machine learning vs statistical methods for time series forecasting: Size matters," 2019, *arXiv:1909.13316*.

- [35] A. Yağmur, M. Kayakuş, and M. Terzioğlu, "Predicting renewable energy production by machine learning methods: The case of Turkey," *Environ. Prog. Sustain. Energy*, vol. 42, no. 3, May 2023, Art. no. e14077.
- [36] U. Şahin, "Projections of Turkey's electricity generation and installed capacity from total renewable and hydro energy using fractional nonlinear grey Bernoulli model and its reduced forms," *Sustain. Prod. Consumption*, vol. 23, pp. 52–62, Jul. 2020.
- [37] S. Çakır, "Renewable energy generation forecasting in Turkey via intuitionistic fuzzy time series approach," *Renew. Energy*, vol. 214, pp. 194–200, Sep. 2023.
- [38] D. B. Unsal, A. Aksoz, S. Oyucu, J. M. Guerrero, and M. Guler, "A comparative study of AI methods on renewable energy prediction for smart grids: Case of Turkey," *Sustainability*, vol. 16, no. 7, p. 2894, Mar. 2024.
- [39] C. Persson, P. Bacher, T. Shiga, and H. Madsen, "Multi-site solar power forecasting using gradient boosted regression trees," *Sol. Energy*, vol. 150, pp. 423–436, Jul. 2017.
- [40] A. Gensler, J. Henze, B. Sick, and N. Raabe, "Deep learning for solar power forecasting—An approach using AutoEncoder and LSTM neural networks," in *Proc. IEEE Int. Conf. Syst. Man, Cybern. (SMC)*, Oct. 2016, pp. 2858–2865.
- [41] M. AlKandari and I. Ahmad, "Solar power generation forecasting using ensemble approach based on deep learning and statistical methods," *Appl. Comput. Informat.*, vol. 20, nos. 3–4, pp. 231–250, Jun. 2024.
- [42] EPIAS. (Apr. 1, 2024). *Injection Quantity*. Accessed: Apr. 1, 2024. [Online]. Available: <https://seffalik.epias.com.tr/electricity/renewables-support-mechanism/licensed-generation-amount/injection-quantity>
- [43] S. Van Buuren and K. Groothuis-Oudshoorn, "mice: Multivariate imputation by chained equations in R," *J. Stat. Softw.*, vol. 45, pp. 1–67, Dec. 2011.
- [44] D. Pyle, *Data Preparation for Data Mining*. San Mateo, CA, USA: Morgan Kaufmann, 1999, pp. 251–252.
- [45] P. Fryzlewicz, "Wild binary segmentation for multiple change-point detection," *Ann. Statist.*, vol. 42, no. 6, pp. 2243–2281, Dec. 2014.
- [46] V. Nair and G. E. Hinton, "Rectified linear units improve restricted Boltzmann machines," in *Proc. 27th Int. Conf. Mach. Learn. (ICML)*, 2010, pp. 807–814.
- [47] S. Hochreiter and J. Schmidhuber, "Long short-term memory," *Neural Comput.*, vol. 9, no. 8, pp. 1735–1780, Nov. 1997.
- [48] J. Bergstra and Y. Bengio, "Random search for hyper-parameter optimization," *J. Mach. Learn. Res.*, vol. 13, no. 2, pp. 281–305, 2012.
- [49] X. Glorot and Y. Bengio, "Understanding the difficulty of training deep feedforward neural networks," in *Proc. 13th Int. Conf. Artif. Intell. Statist.*, 2010, pp. 249–256.
- [50] K. Cho, B. van Merriënboer, C. Gulcehre, D. Bahdanau, F. Bougares, H. Schwenk, and Y. Bengio, "Learning phrase representations using RNN encoder–decoder for statistical machine translation," 2014, *arXiv:1406.1078*.
- [51] EPIAS. *Real-Time Generation*. seffalik.epias.com.tr. Accessed: Apr. 23, 2024. [Online]. Available: <https://seffalik.epias.com.tr/electricity/electricity-generation/ex-post-generation/real-time-generation>
- [52] VisualCrossing. *Historical Weather Data for Turkey*. visualcrossing.com. Accessed: Apr. 29, 2024. [Online]. Available: <https://www.visualcrossing.com/weather-history/turkey>
- [53] Republic of Türkiye Ministry of Health. *COVID-19 Bilgilendirme Platformu*. covid19.saglik.gov.tr. Accessed: Apr. 29, 2024. [Online]. Available: <https://covid19.saglik.gov.tr/TR-66935/genel-koronavirus-tablosu.html>
- [54] Istanbul Metropolitan Municipality. *Open Data Portal*. ibb.gov.tr. Accessed: May 4, 2024. [Online]. Available: <https://data.ibb.gov.tr/en/dataset?q=dam>
- [55] International Monetary Fund. *Real GDP Growth*. imf.org. Accessed: May 2, 2024. [Online]. Available: https://www.imf.org/external/datamapper/NGDP_RPCH@WEO/TUR?zoom=TUR&highlight=TUR
- [56] EPIAS. *Installed Capacity*. seffalik.epias.com.tr. Accessed: Apr. 7, 2024. [Online]. Available: <https://seffalik.epias.com.tr/electricity/renewables-support-mechanism/installed-capacity>
- [57] EPIAS. *Unit Cost*. seffalik.epias.com.tr. Accessed: Apr. 8, 2024. [Online]. Available: <https://seffalik.epias.com.tr/electricity/renewables-support-mechanism/unit-cost>
- [58] A. Kannal, (2020). *Solar Power Generation Data [Dataset]*. Kaggle. Accessed: Mar. 1, 2025. [Online]. Available: <https://www.kaggle.com/datasets/anikannal/solar-power-generation-data/data>
- [59] M. H. Kutner, C. J. Nachtsheim, J. Neter, and W. Li, *Applied Linear Statistical Models*. New York, NY, USA: McGraw-Hill, 2005.
- [60] R. M. O'Brien, "A caution regarding rules of thumb for variance inflation factors," *Qual. Quantity*, vol. 41, no. 5, pp. 673–690, Sep. 2007.
- [61] I. T. Jolliffe and J. Cadima, "Principal component analysis: A review and recent developments," *Phil. Trans. Roy. Soc. A, Math., Phys. Eng. Sci.*, vol. 374, no. 2065, Apr. 2016, Art. no. 20150202.
- [62] I. T. Jolliffe, *Principal Component Analysis*, 2nd ed., New York, NY, USA: Springer, 2002.



YOUSEF ALKHANAFSEH received the B.S. and M.S. degrees from the Electrical Engineering Department, Istanbul Technical University (ITU), in 2020 and 2022, respectively, where he is currently pursuing the Ph.D. degree. He is also a Software Developer, particularly specializing in big data engineering with TurkNet Communication Services Company, İstanbul, Türkiye. His research interests include creating big data clusters, processing large data, utilizing data streaming tools, deploying ML/NN models, designing workflow systems, managing cloud storages, working with databases, implementing containerization, and building pipelines.



TAHIR CETIN AKINCI (Senior Member, IEEE) received the bachelor's degree in electrical engineering, in 2000, and the master's and Ph.D. degrees, in 2005 and 2010, respectively. From 2003 to 2010, he was a Research Assistant with Marmara University, İstanbul, Türkiye. He was a Full Professor with the Electrical Engineering Department, Istanbul Technical University (ITU), from 2020 to February 2024. He also was the Vice Dean with the Graduate School and the Electrical and Electronics Engineering Faculty, from 2020 to 2021. From 2021 to 2024, he was a Visiting Scholar with the University of California Riverside (UCR). Currently, he is an Assistant Project Scientist with the Bourns College of Engineering Center for Environmental Research and Technology (CE-CERT), UCR. His research interests include artificial neural networks, deep learning, machine learning, cognitive systems, signal processing, power systems, renewable energy systems, and data analysis.



ALFREDO A. MARTINEZ-MORALES received the B.S., M.S., and Ph.D. degrees in electrical engineering from the University of California Riverside (UCR), in 2005, 2008, and 2010, respectively. He is currently the Managing Director of the Southern California Research Initiative for Solar Energy (SC-RISE) and holds the position of a Research Professor with the Bourns College of Engineering Center for Environmental Research and Technology (CECERT). He plays a key role as a Principal Investigator with the Sustainable Integrated Grid Initiative (SIGI) and the Distributed Energy Resources Laboratory (DERL), UCR, contributing to the engineering, permitting, and deployment of microgrids throughout Southern California. His current research interests include solar cells, alkali metal ion batteries, highly integrated renewables, energy storage systems, and microgrids.

...